Geo-Insight

A Bayesian approach to ranking
overlooked humanitarian crises

Datathon 2026 — student proof-of-concept

April 2026

Scope. This is a student proof-of-concept produced over a hackathon weekend. It is not a validated operational tool, not peer-reviewed, and not endorsed by any of the data providers referenced. Its rankings are statistical estimates with calibrated uncertainty, and should not be used to allocate humanitarian funding or to substitute for expert judgement. The purpose of the document is to describe what was built, how it was validated, and where it might be useful as a starting point for further work.

Abstract

In 2026, humanitarian agencies asked \$33 B to reach 135 M people; the 2025 appeal raised the lowest amount in a decade, with nineteen country contexts below 50 % coverage. When nearly every response is underfunded, a single coverage ratio stops discriminating. We treat overlookedness as a latent scalar θ per country and infer its posterior from six observed attributes through a hierarchical Bayesian observation model. Attributes are linked to θ through likelihoods matched to their natural supports; slopes are sign-constrained by domain knowledge; partial-data countries are pooled toward the global mean through a population-level prior. Stakeholder preferences are encoded as priors over slopes rather than weights in a sum, so consensus and contested crises are read from posterior overlap. Inference is variational with NUTS calibration on every fit. The candidate pool is the 22 countries with an active humanitarian response plan. Against two independently-curated human benchmarks (CERF UFE and CARE *Breaking the Silence*), the model ties an additive baseline on the two largest benchmark windows and beats it on the smaller windows; a 2024-only fit predicts the March-2025 CERF selection at 4/10 precision @ 10 without seeing 2025 data.

1 Context

The 2026 Global Humanitarian Overview asks \$33 B for 135 M people in need.[1] The 2025 appeal raised \$12 B, the lowest in a decade. Coverage ratios have collapsed: DRC ∼22 %, Yemen and Somalia ∼24 %, Sudan from 69 % (2024) to 35 % (2025). Nineteen contexts now sit below 50 % coverage, up from eight in 2021.

A single coverage ratio cannot discriminate between these contexts. The useful variation is no longer *whether* a crisis is underfunded but *how*, and *for whom*. Three kinds of overlookedness become distinguishable once scoring moves from a scalar to a joint distribution over multiple signals: *forgotten* (low funding, low attention), *contested* (high attention, persistent decline), and *sector-starved* (adequate aggregate, one cluster at 20 %). Each warrants a different response, and current scalar tools conflate them.

Prior work on humanitarian resource allocation using multi-attribute decision methods (e.g. Rye & Aktas [1]) shows that rank orderings depend strongly on the weights chosen by decision-makers.

[1] https://www.unocha.org/publications/report/world/global-humanitarian-overview-2026-enesfr

We take that finding as motivation to model the underlying construct as latent, and to encode stakeholder preferences as priors on how each signal should inform that latent, rather than as weights that determine a score directly.

2 What was built

Six inputs, one latent, four stakeholders, two external validation sources, and a typed semantic layer underneath them all.

Inputs. Five datasets from UN OCHA (HNO, HRP, FTS, CoD-PS, CBPF) and one from the EU Joint Research Centre (INFORM Severity, 67 monthly snapshots consolidated into long format). All inputs are re-downloaded from source on first run; no pre-baked caches are shipped. Two externally-curated lists are used only for validation and never for fitting: CERF Underfunded Emergencies selections and CARE *Breaking the Silence* annual rankings.

Latent. One unobserved scalar θ per country. Higher θ means more overlooked. θ is never observed; it is inferred jointly from the six inputs through the observation model in §3.

Semantic layer. Seventy-one analytic properties are registered in a single specification file, across five derivation levels (L1 raw observations, L2 scale-normalised ratios, L3 inequality indices, L4 windowed trends, L5 the Bayesian posterior). Each property declares its formula, inputs, unit, granularity, and known failure modes. No analytic column exists in code without a declaration.

Surfaces. The result is exposed through a landing page, a methodology walkthrough, a slide deck, and two Streamlit applications (raw-data audit and semantic analysis). Every displayed value traces back to a registered property and, through it, to a source file.

3 Methodology

Unit of analysis is country-year. A country is *eligible* if it has an observed gap per person in need (a_2) — operationally, an active humanitarian response plan with both HNO and FTS entries. In the 2025 cycle this is 22 countries. This matches the population CERF UFE selections are drawn from, so external validation is on the same pool. Countries without an active response plan return no posterior: the construct "overlooked humanitarian crisis" is not well-defined for them.

Six-attribute vector per eligible country:

- a_1 coverage shortfall $= 1 - \min(\text{fund}/\text{req}, 1) \in [0, 1]$
- a_2 gap per PIN $= \max(0, \text{req} - \text{fund})/\text{PIN} \in [0, \infty)$
- a_3 need intensity $= \text{PIN}/\text{population} \in [0, 1]$
- a_4 severity: INFORM 1–5 ordinal
- a_5 donor concentration (Herfindahl over FTS single-destination incoming flows) $\in [0, 1]$
- a_6 intra-crisis equity: PIN-weighted Gini over sector coverage ratios $\in [0, 1]$,

$$a_6(u) = \frac{\sum_{i,j} n_i \, n_j \, |k_i - k_j|}{2 \, (\sum_c n_c)^2 \, \bar{k}_w}, \qquad \bar{k}_w = \frac{\sum_c n_c \, k_c}{\sum_c n_c}.$$

Attributes are left on their natural supports; no min–max normalisation is applied. The observation likelihoods operate on raw scales.

Observation model. For each attribute i, a likelihood matched to its support:

$$a_i(u) \mid \theta(u) \;\sim\; \begin{cases} \mathrm{Beta}\big(\sigma(\alpha_i + \beta_i\theta)\,\phi_i,\ (1-\sigma(\alpha_i+\beta_i\theta))\,\phi_i\big) & i \in \{1,3,5,6\} \\ \mathrm{LogNormal}(\alpha_2 + \beta_2\theta,\ \sigma_2^2) & i = 2 \\ \mathrm{OrderedLogistic}(\alpha_4 + \beta_4\theta,\ \kappa) & i = 4 \end{cases} \tag{1}$$

where $\sigma(\cdot)$ is the logistic function, $\phi_i > 0$ is a per-attribute precision, $\sigma_2 > 0$ is the log-normal scale, and $\kappa = (\kappa_1, \ldots, \kappa_4)$ are the ordered-logistic cut-points. Beta regression respects the bounded support of the four fractional attributes; the log-normal handles the positive per-PIN gap; the ordered logistic handles the qualitative spacing of the INFORM severity steps that an interval-scale treatment would silently distort.

Sign constraints. Every slope is constrained $\beta_i > 0$ via a half-normal prior. This is the operational definition of θ: the latent moves with each attribute the way domain knowledge says it should — higher coverage shortfall, higher per-PIN gap, more concentrated donors, sharper sector imbalance, greater need intensity, and higher severity all correspond to more overlooked. Without sign constraints the model is only weakly identified and the inferred θ picks up arbitrary linear combinations. The location ambiguity ($\theta \mapsto \theta + c$) is resolved by the population-level prior below.

Hierarchical prior on θ.

$$\mu_\theta \sim \mathcal{N}(0,\, 0.3), \qquad \sigma_\theta \sim \text{Half-}\mathcal{N}(0.5), \qquad \theta(u) \mid \mu_\theta, \sigma_\theta \sim \mathcal{N}(\mu_\theta,\, \sigma_\theta^2). \tag{2}$$

Countries with sparse data are pulled toward the population mean by the prior; countries with dense data are barely affected. Sparse-data countries get wider posteriors by construction, not because we down-weight them manually.

Stakeholders as priors over slopes. A stakeholder s is encoded as a prior on the six slopes,

$$\beta_i^{(s)} \sim \text{Half-}\mathcal{N}(m_i^{(s)},\ v_i^{(s)}), \qquad i = 1, \ldots, 6, \tag{3}$$

producing a separate posterior $p(\theta \mid D, s)$ per stakeholder. Four stakeholder priors are pre-registered (CERF, ECHO, USAID, NGO consortia); the weights used are plausible defaults rather than calibrated from allocation history (§8). Consensus and contested crises are read from posterior overlap across stakeholders.

Inference. Variational, with NUTS calibration on every fit. The production guide is a full-covariance Gaussian; a mean-field guide underestimates posterior variance by $\sim$3$\times$ on this model (Spearman $\rho = 0.68$ against NUTS), whereas the full-covariance guide captures the latent-slope correlations and recovers $\rho = 0.89$. Adam, learning rate 3×10^{-3}, 8,000 SVI steps; full run $\sim$3 s on CPU. NUTS reference: 1,000 warmup + 2,000 samples $\times$ 4 chains, $\sim$24 s. Reported credible intervals are SVI estimates; the SVI-vs-NUTS calibration is reported alongside every headline result.

Outputs. Per country: posterior median $\widehat{\theta}(u)$, 5th and 95th posterior percentiles forming a 90 % credible interval, and CI width. Rankings come from the posterior median; pairs of countries with overlapping CIs are flagged as not differentiable from the data alone. A median split on CI width against a median split on sector equity (a_6) produces the four-cell typology in Table 1.

	Narrow posterior	**Wide posterior**
Sectors uneven	consensus sector-starved	contested sector-starved
Sectors even	consensus overlooked	contested balanced

Table 1: Four-cell typology from posterior CI width $\times$ sector Gini.

4 Validation

Two human-curated external benchmarks, neither used in fitting. CERF UFE (the UN Emergency Relief Coordinator's twice-yearly allocations to underfunded crises) indexes an underfunding-and-consensus axis. CARE *Breaking the Silence* (annual top-10 by media attention) indexes an under-reporting axis. Both are produced by methodology independent of anything in this paper.

As an ablation, the Bayesian posterior is compared against a *naïve mean* — the unweighted mean of the six standardised attributes, computed over only the attributes observed for each country. This is the simplest plausible scorer; no latent variable, no uncertainty, no stakeholder priors. The Bayesian result is the proposal; the naïve mean is the floor we aim to clear.

Benchmark	k	**Bayesian**	**Naïve mean**
CERF UFE 2024 w2	10	3/10 (0.30)	3/10 (0.30)
CERF UFE 2025 w1	10	**5/10** (0.50)	**5/10** (0.50)
CERF UFE 2025 w2	7	**2/7** (0.29)	1/7 (0.14)
CARE BTS 2024	10	**3/10** (0.30)	1/10 (0.10)

Table 2: Precision @ k against external benchmarks, HRP-eligible pool ($n = 22$). The Bayesian ranking ties on the two larger CERF windows and beats the naïve mean on the smaller windows and on CARE BTS. CI widths are within 2× of NUTS.

Calibration. The variational posterior is validated against NUTS on every fit. The full-covariance Gaussian guide runs in ∼3 s and recovers NUTS posterior medians at Spearman $\rho = 0.89$.

Posterior predictive checks. For each of the six attributes, 1,000 replicates are drawn from the fitted posterior and compared against the observed marginal. Coverage of the 90 % predictive interval is ≥ 0.91 on every attribute (target 0.90). Per-country Pearson correlation between predicted mean and observed value is high for coverage shortfall (0.76) and donor concentration (0.82) and weak for need intensity, sector Gini, and severity — attributes that sit near their ceilings across the HRP pool (most HRP countries have PIN ≈ population and severity $\in \{4, 5\}$) and carry little cross-country variation for the latent to recover. The latent is best read as a funding-overlookedness axis.

Temporal held-out. A genuine out-of-sample check: fit on the 2024 cycle only (HRP 2024, FTS through end-2024, INFORM through end-2024), then score the 2024-fit ranking against CERF UFE selections made in 2025. The 2024 fit predicts CERF's March-2025 selection at 4/10 precision @ 10 — without ever seeing 2025 data. Year-over-year top-10 overlap is 7/10: HND, HTI, MOZ, SLV, SOM, TCD, VEN persist; ETH, MLI, SYR drop out (their 2024-cycle severity receded); CMR, GTM, NER enter on 2025 evidence.

Benchmark (selection date)	k	**2024-fit (held-out)**	**2025-fit (concurrent)**
CERF UFE 2024 w2 (Dec 2024)	10	**4/10** (0.40)	3/10 (0.30)
CERF UFE 2025 w1 (Mar 2025)	10	4/10 (0.40)	**5/10** (0.50)
CERF UFE 2025 w2 (Dec 2025)	7	2/7 (0.29)	2/7 (0.29)
CARE BTS 2024	10	1/10 (0.10)	3/10 (0.30)

Table 3: Temporal held-out validation. The 2024 fit uses no 2025 data; its precision against CERF UFE selections made in 2025 is a genuine out-of-sample metric.

5 Results

Table 4 lists the ten countries ranked most overlooked in 2025 under the HRP-eligible pool ($n = 22$). Posterior medians are reported with their 90 % credible intervals; pairs of countries with overlapping CIs cannot be reliably ordered from the data alone. The defensible claim is *membership* in this set, not strict ordering within.

#	Country	ISO3	$\widehat{\theta}$	CI low	CI high	c
1	Honduras	HND	−0.107	−0.369	+0.183	1.00
2	El Salvador	SLV	−0.241	−0.513	+0.016	1.00
3	Mozambique	MOZ	−0.252	−0.507	+0.013	1.00
4	Somalia	SOM	−0.262	−0.517	−0.011	1.00
5	Guatemala	GTM	−0.267	−0.554	+0.026	1.00
6	Niger	NER	−0.267	−0.516	−0.015	1.00
7	Haiti	HTI	−0.274	−0.535	−0.010	1.00
8	Cameroon	CMR	−0.286	−0.529	−0.019	1.00
9	Venezuela	VEN	−0.287	−0.526	−0.034	1.00
10	Chad	TCD	−0.304	−0.546	−0.066	1.00

Table 4: Top-10 most overlooked, 2025 (HRP-eligible pool, $n = 22$). Higher $\widehat{\theta}$ is more overlooked; the latent is centred near zero by the population prior. c is the completeness score — fraction of the six attributes observed for the country. All top-10 countries have $c = 1.00$.

6 Where this might be useful

Treat the model as decision support, not automation. Three ways a posterior with calibrated uncertainty and explicit stakeholder lenses might be more useful than a single coverage ratio:

1. Ranking overlapping crises honestly. Across the top-10, 90 % credible intervals overlap substantially. The defensible claim is *membership* in the set, not ordering within. A user who wants an ordered list should know that the order is less supported than the set; the CI-width column makes that explicit.

2. Separating consensus from contested. Five of the 22 HRP-eligible countries sit in the top-10 under all four stakeholder priors (CERF, ECHO, USAID, NGO): HND, SLV, GTM, HTI, SOM. Eleven others appear in the top-10 under some priors but not others. The separation is informative: full- consensus crises are robust to stakeholder preferences; contested crises surface trade-offs that merit a deliberate decision rather than an averaged score.

3. Explaining why a country ranks where it does. Because every attribute is a registered property with declared provenance, the posterior can be decomposed into attribute contributions. The landing page and the analysis app both expose this decomposition: for each top-10 country, which of the six attributes pull θ up, with what magnitudes.

These are uses for which a small POC tool can be a starting point. Anyone adopting the approach seriously would need the extensions listed in §8, independent review, and an operational data pipeline.

7 Limitations

Stated in full.

- **Incompleteness is structural, not imputed.** Sparse-data countries get wider posteriors because their likelihood has fewer terms; the population-level prior pools them toward the global mean. No attribute is ever filled in.

- **The candidate pool is HRP-eligible only ($n = 22$).** Countries without an active humanitarian response plan return no posterior. This is a deliberate scientific choice. Empirically, expansions of the gate (to countries with positive FTS requirements but no HRP, or with $\geq 4/6$ attributes observed) degraded external precision @ k.
- **Stakeholder priors are uncalibrated.** The four pre-registered priors use plausible defaults rather than weights learned from each agency's actual allocation history. The consensus/contested readout is therefore a methodological framework rather than an empirical claim about these four organisations.
- **Variational posteriors may underestimate variance.** A mean-field guide under-covers when latent parameters are correlated; the full-covariance guide adopted here captures those correlations and is calibrated against NUTS on every fit. Reported CI widths remain SVI estimates.
- **Sign constraints are domain-knowledge commitments.** All six slopes are constrained $\beta_i > 0$. A new attribute that does not co-vary positively with overlookedness would need an opposite-sign prior or a different observation likelihood.
- **Cluster-taxonomy drift** between HNO and FTS. A simple normalisation (lower-case + non-alphanumeric $\rightarrow$ underscore) is applied; a curated cluster map is the next step.
- **INFORM methodology rescaling** (February 2026) makes severity indices pre- and post-boundary not directly comparable. Temporal aggregates spanning the boundary use the stable ordinal category, not the continuous severity score.
- **Posterior medians without CIs are misleading.** Every reported ranking carries the CI as a column.
- **Not automation.** The system ranks; humans decide. Every displayed value carries its provenance.
- **External benchmarks are imperfect proxies.** CERF UFE reflects a political process on top of the data; CARE BTS reflects media attention. Agreement with them is evidence; disagreement is not necessarily a failure.

8 What remains to be done

Three extensions, in order of maturity. Items 1 and 2 are planned; item 3 is prototyped and reports measured numbers.

1. Calibrated stakeholder priors. The slope priors in equation 3 are pre-registered with default values. Regressing actual allocation histories on the six attributes would recover *effective* priors for each agency empirically, turning the consensus/contested typology from a methodological commitment into an empirical readout.

2. Temporal AR(1) on θ. Cross-sectional fits at distinct time points (§4) already show stable cross-cycle agreement and useful held-out precision. The natural next layer is a per-country AR(1): $\theta(u,t) = \mu_u + \rho_u(\theta(u,t-1) - \mu_u) + \eta_u(t)$. Annual HRP/FTS cycles give too few time points on their own; INFORM severity is monthly and would anchor a partial-temporal model in which chronic (μ_u) and acute ($\theta - \mu_u$) components fall out of the same posterior.

3. Monthly severity forecast (prototyped). A Mamba-style selective state-space model (two blocks, 19 k parameters) reads 24 months of the INFORM panel — severity plus seven sub-indicators on IPC phase, displacement, access, and fatalities — and forecasts the mean severity H months ahead on the 1–5 scale. The encoder sees no funding data, no stakeholder priors, and no Bayesian posterior; the input is strictly the monthly panel. Temporal split at January 2025.

Held-out MAE is reported against three baselines at each horizon to avoid overclaiming:

H **(months)**	$n_{\mathbf{test}}$	**Mamba**	**Persistence** $y = s_{t-1}$	**Seasonal** $y = s_{t-12}$	**Mamba/Persist**
1	998	0.328	0.306	0.514	0.93×
3	823	0.214	0.195	0.357	0.91×
6	570	0.242	0.217	0.328	0.90×
9	311	**0.204**	0.271	0.319	**1.33**×
12	78	**0.294**	0.407	0.402	**1.38**×

Table 5: Held-out MAE versus forecast horizon. At short horizons the 1–5 severity ordinal moves slowly and persistence ("next year equals today") is unbeatable. At $H \geq 9$ the learned encoder crosses below persistence; at $H = 12$ it leads by 1.38× on the 78 held-out windows. We do not claim uplift at $H < 9$.

The per-country $H = 12$ forecast is registered as a Level-4 property and can be consumed by the Bayesian model as a seventh sign-constrained attribute. On the 22-country pool the 6-to-7-attribute refit shifts the top-10 by three countries with no net precision gain across the four external benchmarks. The forecasting result above is the standalone contribution; the integration into the Bayesian model is extension, not uplift.

9 Summary

The project treats overlookedness as a latent scalar and infers its posterior from six observations through a hierarchical Bayesian model. Every ranking carries a 90 % credible interval. Stakeholder preferences are encoded as priors over slopes rather than weights in a sum. The variational posterior is calibrated against NUTS on every fit. Against two independent human-curated benchmarks the model ties an additive baseline on the larger windows and beats it on the smaller windows and on the media-attention benchmark; a 2024-only fit predicts the March-2025 CERF selection at 4/10 precision @ 10. The tool is a student proof-of-concept; anything operational would need the extensions above, independent review, and an operational data pipeline.

References

[1] Rye, C. S. & Aktas, E. (2022). *A decision-support framework for the allocation of humanitarian aid supplier contracts using MADM and cluster analysis.* International Journal of Disaster Risk Reduction, 78, 103138.